Yeming Dai

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RESEARCH INTERESTS

Reliable LLM agents; verification and evaluation; human oversight and state tracking in long-running agent workflows; robust machine learning.

EDUCATION

Arizona State University | Jan 2024-Dec 2026 (expected)

B.S. in Computer Science (Software Engineering) | GPA: 4.0/4.0

Northern Arizona University | Non-degree physics coursework | Dec 2025-Jan 2026

Anhui University of Science and Technology | Sep 2021-Jul 2023

Software Engineering undergraduate studies; no degree awarded

PUBLICATIONS AND MANUSCRIPTS

Adrian Ciotinga, **Yeming Dai**, and YooJung Choi. *PeTeR: Post-Training Robustification of Probabilistic Circuits.*

Workshop paper: UAI 2026 Workshop on Tractable Probabilistic Modeling (published). **Extended manuscript:** AISTATS 2027 (under review).

RESEARCH EXPERIENCE

Undergraduate Researcher in Probabilistic Models

Arizona State University | *Mar 2026-Present*

Research on post-training robustness of probabilistic circuits under distribution shift.

- Identified probability-normalization drift under direct optimizer updates; validated post-update clamping and renormalization while preserving gradient propagation.
- Investigated a label-alignment confound in random-embedding evaluation; repeated the analysis with fixed labels and ruled out label mismatch as the cause of poor clustering.
- Executed and audited a PeTeR sweep covering 3,360 hyperparameter configurations, recording numerical failures at high learning rates.
- Ran and audited GPU experiments across 28 datasets; reconstructed the CUDA/PyTorch/Triton/PyJuice runtime and selected concurrency from measured GPU usage.
- Connected training and evaluation across two codebases through compatible model serialization; built a reviewer-facing runner with packaged datasets and CSV export.

INDEPENDENT SYSTEMS WORK

LLM Agent Workflows and Independent Review

Independent project | *2026* | Method | Prototype | Technical note (non-peer-reviewed)

- Separated stable requirements from changing task state, but found that future plans still depended on execution results. Distinguished accepted results, the current task, and conditional next steps so plans could be revised as evidence changed.
- Built a runnable two-agent prototype with separate execution and review contexts, evidence-linked decisions, checkpoint/resume support, and human approval before final acceptance.

LLM Agent Workflows and Independent Review continued

- Applied the prototype to multilingual support in an existing transcription application, producing traceable commits and a reviewed handoff for live Japanese and French transcription checks.
- In a packaging task, independent agent review caught a symbolic-link loophole missed by passing tests and required corrected directory checks.
- Compared candidate claims with prior work and narrowed subsequent questions to verification behavior and human decision boundaries.

SELECTED TECHNICAL PROJECTS

Classroom Live Transcriber

Independent developer | *May-Aug 2026* | Repository

- Built and released an on-device Apple Silicon lecture-transcription application using Metal-accelerated whisper.cpp, overlapping audio chunks, resampling, and asynchronous workers.
- Preserved raw and cleaned transcripts, logs, and session configuration; implemented model size/SHA-256 verification, atomic publication, retries, and failure-safe downloads.
- Validated the packaged application across multiple Macs through model setup, real transcription, and Stop/Restart workflows; observed over one hour of classroom use without crashes or sustained backlog.

AudioShifter

Independent developer | *Mar 2025-Aug 2026* | Repository

- Independently developed the original Windows application and an Apple Silicon macOS version; delivered the tested, packaged macOS pre-release within 3 days in August 2026.
- Implemented a modular architecture with background processing, cancellation, child-process cleanup, and source-file protection.
- Verified the pre-release with 165 automated tests, including fault injection, real audio pipelines, packaged execution, and GUI behavior; recorded no failures or skips.
- Audited 75 native binaries and 324 dynamic-library references, then validated the packaged application on a separate non-development Mac.

HONORS AND SCHOLARSHIPS

New American University Transfer Scholarship, ASU (four semesters).

Dean's List, ASU: Spring 2024, Fall 2024, Spring 2025, Fall 2025.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, C/C++, Bash/Shell

ML and research computing: PyTorch, CUDA, Triton, PyJuice, NumPy, Slurm

Systems and tooling: Linux, macOS, Git, pytest, Metal, whisper.cpp, PySide6, FFmpeg, PyInstaller